

Paper Notes

Section 1

- General Conservation Equation:

$$\partial_t F + v \cdot \nabla_x F + \mathcal{F} = C \quad (1)$$

where $F = F(t, x, v)$, $\mathcal{F} = \mathcal{F}(F, \nabla_v F)$,
and $C = C(F)$.

- LHS of (1) is the streaming operator that models particle dynamics in phase-space.

- RHS of (1) is the collision operator that models particle dynamics due to collision in velocity-space

- (1) is difficult to solve because:

- high dimensionality
- complex + multiscale operator

- Low-rank methods are grid-based

- do not suffer from ~~the~~ noise

Section 2

- Ingredients to reduce complexity For dynamical systems
 - reduce the amount of DoF (reduce memory)
 - computational efficiency

- Main example

$$F(t, x, v) = \sum_{i,j=1}^r U_i(t, x) S_{ij}(t) V_j(t, v) \quad (2)$$

where r is the very small rank.

- For the 3D setting, memory consumption goes from $O(n^6)$ to $O(rn^3)$.
- Although reminiscent of SVD, $S(t)$ in (2) may be a full $n \times n$ matrix.
- Why do most kinetic problems admit small r ?
- Examples of low-rank models

- diffusive limit

- radiative transfer

$$F(t, x, v) = p(t, x) + \varepsilon g(t, x, v)$$

where ε is collisionality strength and p solves parabolic diffusion eq

- Fluid limit

- Maxwellian

$$F(t, x, v) = \exp\left[-\frac{(v - u(t, x))^2}{2\theta(t, x)}\right] (p(t, x) + \varepsilon g(t, x, v))$$

- Weakly nonlinear collisionless

- $\partial_t F + v \cdot \nabla_x F = 0$

- + $F(0, x, v) = e^{ikx} e^{-\frac{v^2}{2}}$

- then $F(t, x, v) = e^{ikx} e^{-(i k \cdot v + \frac{v^2}{2})}$

- While F is a rank-1 solution, it's hard ~~for~~ numerically to catch the high-dimensionality. The reason is that longer physical time creates smaller structures in phase space, requiring more grid points.

- Where $F = U, S, V$, since V and U are rank-1 the order of complexity $O(n_x^3 + n_v^3)$ (or $O(n^3)$ if $n_x = n_v$) compared to $O(n_x^3 n_v^3)$ (or $O(n^6)$ if $n_x = n_v$).

- For small fields $E + B$ From the Vlasov-Maxwell equation, $\partial_t F + v \cdot \nabla_x F - (E + B \times v) \cdot \nabla_v F = 0$, solutions behave in similar ways.

- Other low-rank methods

- Hermite method

- $F(t, x, v) = \sum_{i=1} U_i(t, x) H_i(v)$

- Fixes a basis for velocity through time
 - requires a large basis to capture sol.
 - good when near fluid limit

Section 3

- DLR

• 1D VP Eq

$$\partial_t F = -v \partial_x F + E \partial_v F =: \text{RHS}(F) \quad (4)$$

$$E(x,t) = -\partial_x \Phi, \quad -\partial_{xx} \Phi = \rho, \quad \rho = 1 - \int F dv$$

• Projection where $F = \sum_{ij} u_i s_{ij} v_j$

$$P(F)g = P_u g + P_v g - P_u P_v g \quad (5)$$

• Lemma 4.1

$$\partial_t F + P(F)[v \partial_x F - E \partial_v F] = 0$$

• Prop 2.1 ($\text{RHS}_{LR} := \text{RHS}(\sum_{ij} u_i s_{ij} v_j)$)

$$\partial_t s_{ij} = \langle u_i v_j, \text{RHS}_{LR} \rangle_{xv} \quad (6)$$

$$\sum_j s_{ij} \partial_t v_j = \langle u_i, \text{RHS}_{LR} \rangle_x - \sum_j \partial_t s_{ij} v_j \quad (7)$$

$$\sum_i s_{ij} \partial_t u_i = \langle v_j, \text{RHS}_{LR} \rangle_v - \sum_i \partial_t s_{ij} u_i \quad (8)$$

• Projector Splitting Integrator $\mathcal{E}'$

◦ Uses a semi-Lagrangian approach

$$- \partial_t F = P_u \text{RHS}(F)$$

$$- \partial_t F = P_v \text{RHS}(F)$$

$$- \partial_t F = -P_{uv} \text{RHS}(F)$$

◦ Define

$$- K_j = \sum_i u_i s_{ij}$$

$$- L_i = \sum_j s_{ij} v_j$$

◦ Algo

$$- \text{Solve } K^{n+1} w / \partial_t K \text{ (Forward)} \rightarrow \boxed{U^{n+1} + S^{*}} \leftarrow$$

$$- \text{Solve } S^{**} w / \partial_t S \text{ (backward)} \rightarrow S^{**} \quad \text{QR}$$

$$- \text{Solve } L^{n+1} w / \partial_t L \text{ (Forward)} \rightarrow \boxed{V^{n+1} + S^{n+1}} \leftarrow$$

◦ Problems change the CFL condition.

- Basis update + Galerkin Integrator (BUG)
 - Similar Semi-Lagrangian Approach
 - Updates K^n and L^n independently to only extract U^{n+1} and V^{n+1} to then use to update S^n (all forward in time)
 - However, this introduces errors since S^n uses future bases ($U^{n+1} + V^{n+1}$). ~~This~~
 - To remedy this, create augmented bases $\hat{U} + \hat{V}$ from QR decomp of matrix concatenations $[K^{n+1}, U^n]$ and $[L^{n+1}, V^n]$, respectively. Then we use $\hat{U} + \hat{V}$ to update S^n .
 - However, this creates an S^{n+1} in $\mathbb{R}^{2r \times 2r}$, so we perform SVD on $S^{n+1} \approx S$, truncate to a tolerable rank- $\tilde{r}$, then update the following $S^{n+1} = \sum_{\tilde{r}} \hat{U}_i^{n+1} (P_{ki})_{\tilde{r}}$ and $\hat{V}_j^{n+1} = \sum_{\tilde{r}} \hat{V}_l (Q_{lj})_{\tilde{r}}$, where $S = P \Sigma Q^T \approx P_{\tilde{r}} \Sigma_{\tilde{r}} Q_{\tilde{r}}^T$.
 - In [28], they found a parallelized version that needs not the rank- $2r$ update.

- All DLR methods are first order accurate in time, while the splitter can become better w/composition. Strang splitter is 2nd-order. Parallel Bug is 2nd-order.

- Rank-adaptive methods improve models

Section 4

- Domain $(x, v) \in \Omega = \Omega_x \times \Omega_v$

- Tensor Form

$$F = \sum_{ij} s_{ij} u_i \otimes v_j$$

- Flux-difference conserves mass

- Upwind schemes are used for stability

- DLR has issues w/BCs (see [85, 99])

- Semi-Lagrangian methods take advantage of the VP eq's constant sol. on characteristic curves

- In [100], it was shown through classic von Neumann linear stability analysis can apply to steps K, S, + L.

- This is true because splitters and BUGs approximate the nonlinear evolution eqs for low-rank factors w/ a sequence of linear steps, then interposed by QR, SVD, and truncation. All of which are L^2 stable.

- DLR method types

- discr. the problem then derive the syst. of ODEs to solve (D-t-O)

- derive low-rank equations for the cont. syst then discr. in space/velocity (O-t-D)

- BCGs achieve L^2 stability under the CFL condition in both cases.
- Splitters in D-t-O schemes don't play well with numerical diffusion but do have promise for stability in O-t-D [100].

- Consider the S-step w/ explicit Euler

$$\bullet S_{ij}^{n+1} = S_{ij}^n + \Delta t \langle \text{RHS}^{(A)}(\hat{U}, S^n, \hat{V}), \hat{U}_i \hat{V}_j \rangle_{\Delta} \quad (26)$$

where each " Δ " denotes discretization (first)

$\bullet$ Let $F_{\Delta}^m := \hat{U} S^m \hat{V}^T$ then

$$\text{RHS}^{(A)}(\hat{U}, S^n, \hat{V}) = \hat{U}^T \text{RHS}^{(A)}(F_{\Delta}^n) \hat{V}.$$

$\bullet$ Subbing these into (26), we get

$$\begin{aligned} (\hat{U} F_{\Delta}^{n+1} \hat{V})^T &= \hat{U}^T (F_{\Delta}^n + \Delta t \text{RHS}^{(A)}(F_{\Delta}^n)) \hat{V} \\ \Rightarrow F_{\Delta}^{n+1} &= \hat{U} \hat{U}^T (F_{\Delta}^n + \Delta t \text{RHS}^{(A)}(F_{\Delta}^n)) \hat{V} \hat{V}^T \end{aligned}$$

$\bullet$ By Frobenius norm

$$\|F_{\Delta}^{n+1}\|_F \leq \|F_{\Delta}^n + \Delta t \text{RHS}^{(A)}(F_{\Delta}^n)\|_F$$

- Check [102, 50, 12, 126] for other derivations of stability for diff schemes/problems.

Section 8

- Open Problems For DLRA

- Derivation of robust error bounds
 - Current bounds limited to Lipschitz cont. RHS (i.e. non-stiff) ([88])
- possible approaches [
 - Show DLRA is smooth to use ~~the~~ Taylor Series expansion
 - Show stability w/ general inputs w/ smooth IC
 - It's assumed that the RHS ~~for~~ dynamics push the computations outside the low-rank manifold by a tiny amount, motivated by Lipschitz cont. by not shown For general RHSs
- ? [
 - No 2nd order robust error bound has been found for the projector-splitting integrator
 - Non-Lipschitz RHSs (i.e. stiff) have only L^2 stability and energy stability results
 - [161] shows DLR converges to equilibrium w/ the omission of stiff terms
- Domain Decomp
 - Solve independent / low-dependent parts of the domain in parallel
 - Local low-rank decoms for subdomains (Chand)
 - How to transfer low-rank data on B w/ numerical fluxes and/or ghost regions?

- Implicit Integrators (Low-Rank)
 - transport sweeps work w/ low rank
 - methods on structured grids [130]
- Positivity Preservation & ~~Low~~ Low-rank Hamiltonian Preservation
- Solving the Boltzmann-BGK Operator
- Sparse & Low-Rank sorting